

1.

Machine Learning is used in AI for making computers learn automatically. It works using data and gives output.

There are types of Machine Learning:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

Supervised learning uses labeled data. Unsupervised learning finds patterns. Reinforcement learning uses rewards.

Machine Learning is used in many apps and websites nowadays.

2.

Overfitting means the model learns too much and does not work properly for new data.

To avoid overfitting:

- Use more data
- Stop training
- Use dropout
- Remove extra data

Overfitting is bad because accuracy can decrease on testing data.

3.

Reinforcement Learning is learning using rewards and punishments. The machine learns from experience.

If the action is correct, reward is given. If wrong, punishment is given. Slowly the machine improves.

Uses are:

- Games
- Robots
- Cars
- AI systems

Self-driving cars use Reinforcement Learning for driving. It is important in Artificial Intelligence.

4.

Data preprocessing means cleaning the data before giving to the model.

Sometimes data has missing values and mistakes. Preprocessing removes them.

Good data gives good output. Bad data gives bad output.

The model does not behave same for both good and bad data. If data is wrong, prediction also becomes wrong.

So preprocessing is important in Machine Learning projects.

5.

NLP means Natural Language Processing. It is used for understanding human language by computer.

Examples:

- Chatbot
- Google Translate
- Voice systems

Embeddings are important because words are converted into numbers. Machine understands numbers better.

Embeddings help find similar words and improve NLP models.

6.

Machine Learning is used when there is more data and prediction problems.

It is used in:

- YouTube recommendations
- AI
- Image detection

Machine Learning should not be used for simple calculations because normal programs are enough.

ML also needs data and training. Without good data, results may become wrong.